

Clinical Risk Assessment Report: Patient 3

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 45.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.07x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +6.7%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.52 [Bottom <1%] (+6.7%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.55 [Top 7%] (+4.3%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.36 [Top 8%] (+3.4%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.44 [Top <1%] (+3.1%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.67 [Avg (67th %ile)] (+3.0%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.6 [Bottom 18%] (+2.8%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.28 [Bottom <1%] (+2.3%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 2.01 [Top <1%] (+1.1%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

